Laith Assaf

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EDUCATION

Michigan State University, College of Engineering Expected May 2027
BS Computer Science, Concentration in AI/ML | Minor in Economics East Lansing, MI

EXPERIENCE

PC Technician Intern May 2026 - Present
Onsite Support Team, Delta Dental Okemos, MI

- Built LLM agents in **Copilot Studio** and AI Builder across 3 Power Automate workflows, converting natural-language requests into structured records using live Ivanti asset and user data retrieved via API
- Shipped an asset management workflow to production, saving 5 minutes per run and **$11K annually**
- Shipped an agent that parses incoming emails and updates the matching tickets, rolled out across IT teams
- Imaged and configured 100+ endpoints through Ivanti, enrolling each device in Active Directory and Intune so it reached its user with policy applied, and backing up and wiping each drive at the end of the device lifecycle
- Resolved 100+ tickets spanning laptop upgrades and hardware faults, restoring locked-out users through Azure BitLocker recovery and CyberArk credential access, and tracing network port issues with Cisco DDI

IT and Media Services Support Associate May 2023 - May 2026
MSU College of Osteopathic Medicine East Lansing, MI

- Supported 100+ classroom AV and conferencing systems across 4 buildings serving 1,200+ students
- Resolved 50-70 tickets weekly at a 5-10 minute average response time, maintaining **99.5% uptime** during live lectures through a documented troubleshooting process for network, display, audio routing, and Zoom failures

PROJECTS

Real-Time Audio Classification | *Python, TFLite, MemryX, Flask* | **2nd Place, MHacks** September 2025

- Placed 2nd of 300+ participants at MHacks 2025 with a real-time audio classifier that alerts hearing-impaired users to surrounding sounds, reaching **95% accuracy** at <50ms latency across 500+ sound categories
- Ran YAMNet on-device on a Raspberry Pi 5 with a MemryX MXA accelerator, slicing 16 kHz microphone audio into 1-second windows and feeding the MXA through asynchronous callbacks so capture never blocks inference
- Filtered 521 sound classes down to 11 safety sounds (fire alarms, sirens, doorbells, baby cries) in three priority tiers with a 5-second cooldown, pushing alerts to a live dashboard over Flask-SocketIO and to Discord by webhook

Multiplayer Party Game Framework | *TypeScript, Socket.io, Next.js* December 2025 - April 2026

- Designed and published the framework as three open-source TypeScript packages on npm, packaging its game state machine, WebSocket rooms, and player lifecycle into a 5-hook API with **80%+ code reuse** across new games
- Architected a multi-room orchestration system on Express and Socket.io namespaces with Firebase Realtime DB state sync, hosting 4 concurrent games on a single Railway server with TTL-based room cleanup
- Kept the framework and its consumer game repos in sync through a documented issue-flow contract

Raspberry Pi TV Home Screen | *Rust, Slint, Linux, Wayland* October 2026 - Present

- Building an Apple TV-style home screen for a Raspberry Pi 5 media box in Rust that launches and switches between Jellyfin, Steam Link, and Kodi, with a Home key that returns to it from inside any app
- Moved the UI from WebKit to Slint after benchmarks showed Slint held **60 fps** where WebKit ran 9.8% slow frames, then raised 4K output from 27.8 to 59.7 fps by pre-rendering gradients and shadows on the CPU
- Gated each milestone on criteria set before implementation and measured on the Pi: Home returns in under 10 ms, a killed app hands focus back within 41 ms, and 120 automated tests pass

LEADERSHIP

Michigan State University | AI Club January 2025 - Present
Workshop Coordinator & Project Lead East Lansing, MI

- Delivered 8+ workshops on local LLM deployment (Ollama, LM Studio), agent orchestration, and PyTorch to **500+ attendees** at 90% retention and 4.8/5 satisfaction; led a 4-student team building an LLM course predictor

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, C++, C#, SQL
Web & Backend: Node.js, Express, React, Next.js, FastAPI, Flask, Socket.io, REST APIs, WebSockets, Firebase
Infrastructure: Docker, Linux, Git, GitHub Actions CI, PostgreSQL, Railway, Vercel
AI & Automation: LLM APIs, agent orchestration, MCP, Copilot Studio, Power Automate, PyTorch